

ECON1010

Introductory Microeconomics

Tutorial 9 | Imperfect Competition – Strategy



This Week in ECON1010

CML 4 is due **Tuesday** 4.00pm May 05

CML 5 opens **this week** 9.00am May 06

Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- ✉ Email
- ☎ 1300 275 870 (8.30am-5pm)
- 📍 Building 42, St Lucia
- 📍 Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

UQ Campus Security
☎ 07 3365 3333 (24/7)

Off-campus emergency
☎ 000 (24/7)

Crisis support

For urgent mental health support:

UQ Counselling and Crisis Line
☎ 1300 851 998 (24/7)

Text a Crisis Counsellor
☎ 0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

Imperfect Competition – Strategy

Market Structures

Increasing Competition



**Perfect
Competition**

**Monopolistic
Competition**

Oligopoly

Monopoly

Increasing Concentration



Question 1a

In your own words what is:

- i) First degree price discrimination
- ii) Second degree price discrimination
- iii) Third degree price discrimination
- iv) Peak load pricing



Question 1a

ii) First degree price discrimination

First degree price discrimination is considered to be ‘perfect’ discrimination, as the firm knows the maximum each customer will pay and charges each consumer this price. The firm captures each consumer’s full desire to pay, resulting in 100% supplier surplus and 0% consumer surplus.

Example: A second hand car salesman negotiates a different price with every new customer to reach their highest possible price.

ii) Second degree price discrimination

Second degree price discrimination is a difference price per quantity of products purchased. Example: “Buy 1 for \$10 or 2 for \$15.”

Example: Boost Juice - Small \$6.30, Medium \$7.30, Large \$7.70

Question 1a

iii) Third degree price discrimination

Third degree price discrimination differentiates customer groups (market segments) based on their price elasticity of demand. Customer groups who have less money are charged lower prices.

**Example: Students and Seniors discounts to waterparks and movies.
Example: Expensive flights arriving into major cities on Monday mornings and departing Friday afternoons. As business professionals are more likely to fly at these times, the airlines may charge higher prices.**

iv) Peak load pricing

Peak load pricing refers to cost and price increases when a product / service is in high demand.

Example: Translink (Brisbane City public transport) charges 20% extra during peak hours 6:00am – 8:30am and 3:30pm – 7:00pm. The extra 20% reflects the extra energy costs(extra weight on busses, trains and water ferries), extra vehicles required to be in operation and extra security at the terminals.

Note: this example also appears to be third degree price discrimination as this time period is when students and many professionals must travel to school and work (they therefore have a lower price elasticity of demand than other travellers).

Question 1b

A key feature of game theory is the idea of strategic interdependence - the actions of one firm having major impacts on the actions of other firms in the same market. Which of the following circumstances do not involve game theory? Explain why or why not.

- i) A BP petrol station owner wondering how the 7/11 petrol station owner across the road will react if BP decides to lower the price of their unleaded petrol.
- ii) Negotiating your salary when two potential employers have made you a job offer.
- iii) A firm's behaviour in a perfectly competitive market.
- iv) Playing the card game poker.



Game theory is where decision makers (firms or individuals) are influenced by the decisions of a competing decision maker. This is known as interdependence.



Question 1b

i) A BP petrol station owner wondering how the 7/11 petrol station owner across the road will react if BP decides to lower the price of their unleaded petrol.

This does involve game theory as BP lowering their prices to attract 7/11's customers will push 7/11 to decrease their prices as well. The result will be a loss for both petrol station owners.

ii) Negotiating your salary when two potential employers have made you a job offer.

This does involve game theory as one employer is likely to make a better offer if their competitor is offering better work conditions.

iii) A firm's behaviour in a perfectly competitive market.

This does not involve game theory. Firms in perfectly competitive markets are price takers and cannot influence the market price or conditions. Because the firms are already operating at marginal cost, it would be very costly if they lowered their prices to attract more sales.

iv) Playing the card game poker.

This does involve game theory. Poker players are interdependent because one player's action (bet, bluff, fold) directly impacted the decision making of their opponents.

Question 2a

Two rival pizza shop-owners, **Mario** and **Luigi**, are working out their own pricing strategies. At the moment, they both charge a \$3 surcharge for home-delivery and receive a total of \$4000 per week in revenue from their home-delivery pizzas.

Mario is considering cutting his home delivery surcharge to \$2.50, which would increase his sales and weekly revenue to \$5000 while causing Luigi's to fall to \$3000, as long as Luigi keeps his \$3 surcharge.

Across the street, Luigi is also considering cutting his surcharge to \$2.50, which would increase his weekly revenue to \$5000 while causing Mario's to fall to \$3000, as long as Mario keeps a \$3 surcharge.

If both Mario and Luigi decide to cut their home delivery surcharge to \$2.50, they can both expect to earn \$3500 in weekly revenue.

i) Enter the relevant information in the payoff matrix below

		Mario	
		Keep \$3 surcharge	Cut surcharge to \$2.50
Luigi	Keep \$3 Surcharge		
	Cut surcharge to \$2.50		

ii) Luigi's decision to cut his surcharge to \$2.50 would be:

- a. a dominant strategy
- b. cost maximising
- c. a profit maximising strategy
- d. both (a) and (c)
- e. none of the above

iii) If the pizza shop owners are unable to cooperate, what will be the equilibrium outcome?



Question 2a

i) Enter the relevant information in the payoff matrix below

		Mario	
		Keep \$3 surcharge	Cut surcharge to \$2.50
Luigi	Keep \$3 Surcharge	\$4000, \$4000	\$3000, \$5000
	Cut surcharge to \$2.50	\$5000, \$3000	\$3500, \$3500

ii) Luigi's decision to cut his surcharge to \$2.50 would be:

A dominant strategy results in a better payoff regardless of the competitor's action. As all situations are better off if Luigi cuts his surcharge to \$2.50, this is his dominant strategy.

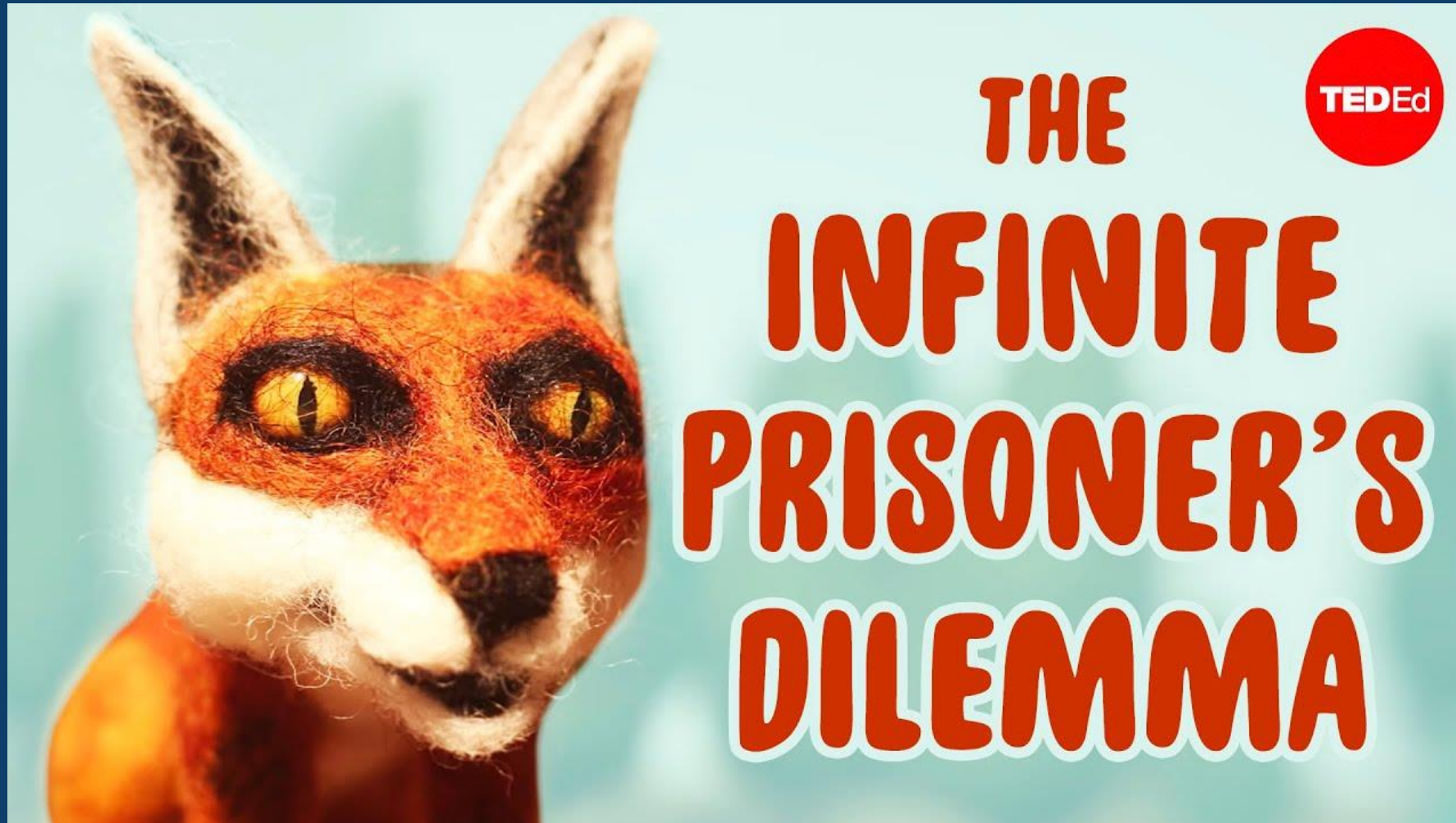
Note: no mention of costs and we cannot tell if it is a profit max strategy without MR and MC

iii) If the pizza shop owners are unable to cooperate, what will be the equilibrium outcome?

As Mario and Luigi both have the same dominant strategy (cut surcharge to \$2.50), both will reduce their price resulting in the worse payoff (\$3500, \$3500). This is a typical Prisoner Dilemma scenario. If Mario and Luigi cooperate and choose to keep the same surcharge price instead, they will both earn higher revenues (\$4000, \$4000). This is why cartels (suppliers that cooperate in price-setting) are profitable.

Note: The most profitable outcome (\$4000, \$4000) is not a Nash equilibrium as both players have an incentive to deviate to (cut surcharge to \$2.50). In order to secure the (\$4000, \$4000) outcome, by preventing the players from deviating, there must be some hidden commitment or motivation in force (anti-competition contracts, threats etc) which are normally illegal.

Quick Intermission – Prisoner's Dilemma Video



<https://www.youtube.com/watch?v=emyi4z-O0ls>

Question 2b

You and your friend decide to work together during this course to produce study notes for the final exam. You both agree to write down notes during the weekly live lectures, then share them with each other afterwards. You soon begin to realise after a few live lectures that you actually learn new concepts better by just sitting and listening (without writing notes), then later reviewing your friend's notes. Therefore, you must decide whether you take and share notes or don't take notes (and sit and listen).

		FRIEND	
		Don't take notes	Take and share notes
YOU	Don't take notes	60%, 60%	95%, 80%
	Take and share notes	80%, 95%	85%, 85%

- i) Is there a dominant strategy for either you or your friend?
- ii) Is it a credible threat when your friend says "If you don't take notes, I won't take notes either"?
- iii) Assume you and your friend are able to change your decision in response to the other's decision (ie a game scenario with a time dimension). Draw a decision tree where you make the first move.
- iv) Draw another decision tree where your friend makes the first move.
- v) Considering the two possible decision trees, what is the Nash Equilibrium?
- vi) Assume you and your friend decide at the same time (ie a simultaneous game) and must then stay with this decision, what is the Nash Equilibrium?



Question 2b

i) *Is there a dominant strategy for either you or your friend?*

No. Both players' best decision depends on the decision of the other player. If your friend doesn't take notes, you would choose to take and share notes (and get 80% rather than 60%). However, if your friend takes and shares notes, you would choose not to take notes (and get 95% rather than 85%).

ii) *Is it a credible threat when your friend says “If you don't take notes, I won't take notes either”?*

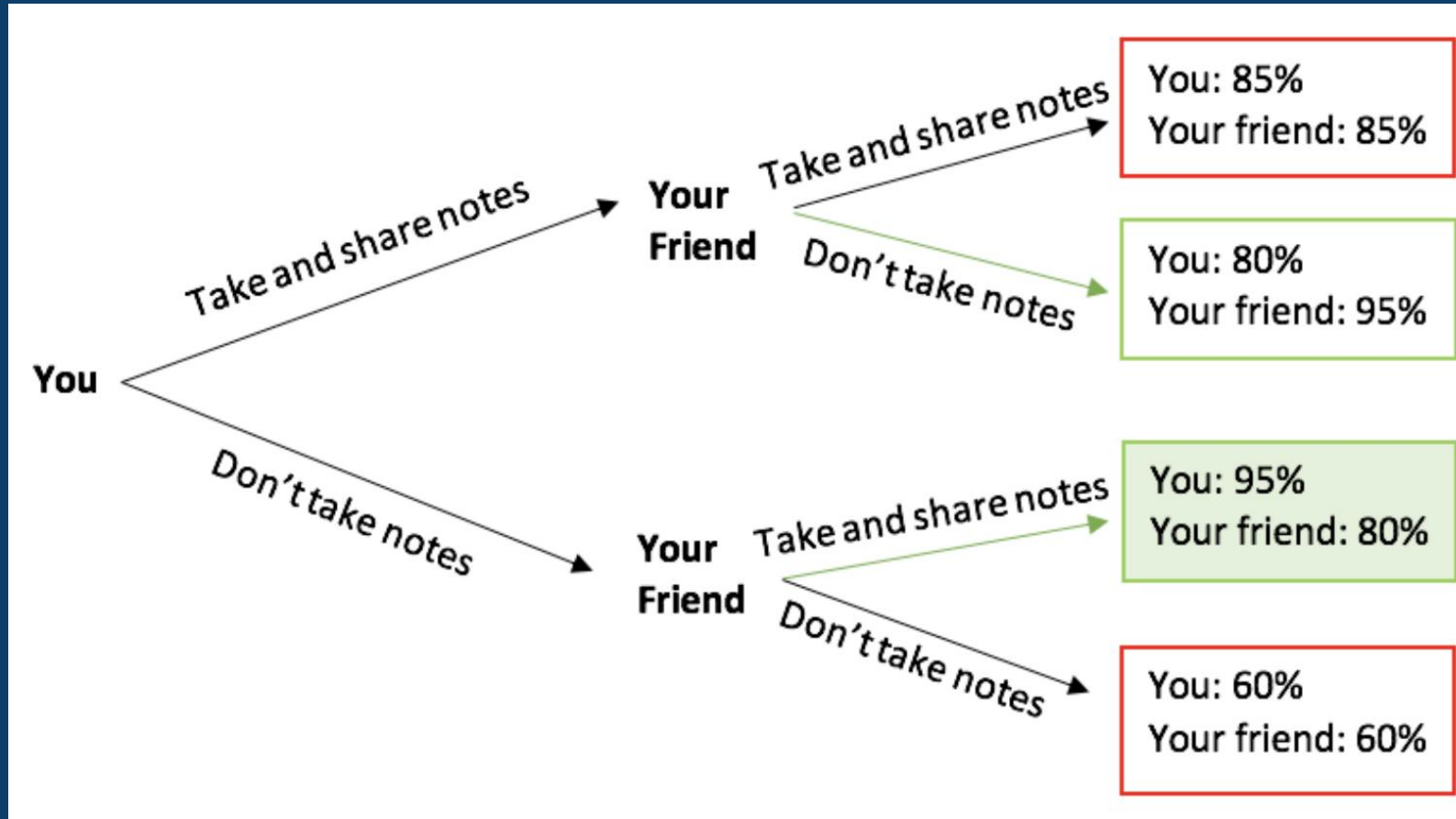
No. It is not a credible threat because it is not in your friend's interest to not take notes because this will reduce their own grade down to 60%.

		FRIEND	
		Don't take notes	Take and share notes
YOU	Don't take notes	60%, 60%	95%, 80%
	Take and share notes	80%, 95%	85%, 85%



Question 2b

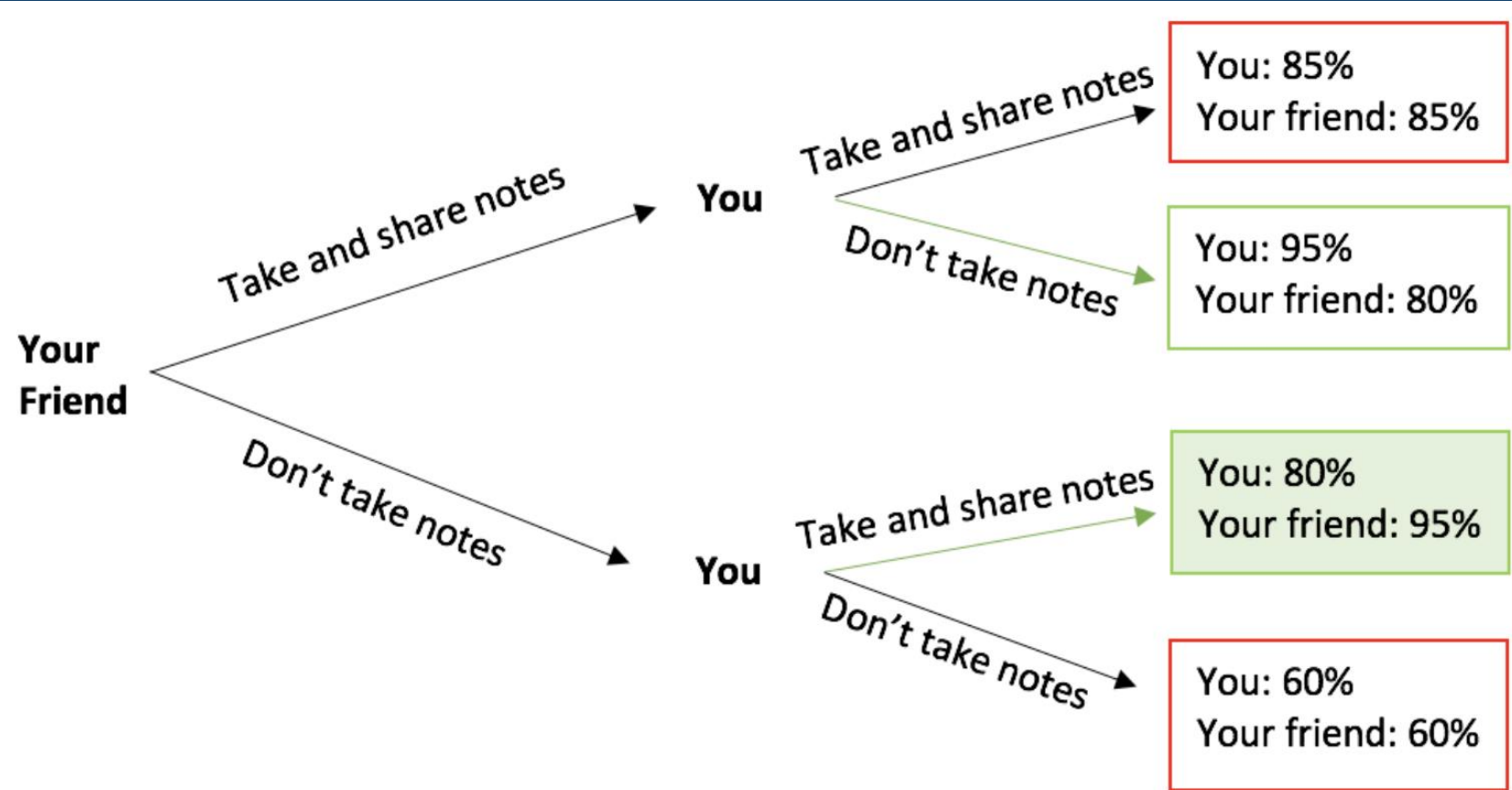
iii) Assume you and your friend are able to change your decision in response to the other's decision (ie a game scenario with a time dimension). Draw a decision tree where you make the first move.



When you go first, the green boxes (Take and share notes, Don't take notes) and (Don't take notes, Take and share notes) are the only possible final outcomes – your friend will not choose the red boxes as this reduces their grade. The decision (strategy) that maximises your grade is Don't take notes. Your friend will then choose Take and share notes to maximise their grade.

Question 2b

iv) Draw another decision tree where your friend makes the first move.



The scenario is the same for your friend. The decision (strategy) that maximises their grade is Don't take notes. You will then choose Take and share notes to maximise your grade.

Question 2b

v) Considering the two possible decision trees, what is the Nash Equilibrium?

If the game includes a time dimension, whichever player goes first will choose not to take notes and the other will choose Take and share notes to maximise their final grade. The outcome will therefore be:

- Your friend goes first: (Take and share notes, Don't take notes) = (80%, 95%)
- You go first: (Don't take notes, Take and share notes) = (95%, 80%)

This is the Nash Equilibrium because, once reached, no player can change their decision to better their outcome.

vi) Assume you and your friend decide at the same time (ie a simultaneous game) and must then stay with this decision, what is the Nash Equilibrium?

Remember: the Nash Equilibrium is where neither player can benefit from changing their decision if their opposition's decision remains the same. As it is better here for both friends (players) to choose different actions (opposing strategies) and it is detrimental for either to change (deviate) from these positions, the Nash Equilibria are therefore:

		FRIEND	
		Don't take notes	Take and share notes
YOU	Don't take notes	60%, 60%	95%, 80%
	Take and share notes	80%, 95%	85%, 85%

Question 3

Work through the following to develop your understanding of how game theory is present in your daily life.

- i) List three different activities you and a friend could do this Friday evening.
- ii) Estimate (in dollar amounts) how much happiness each activity gives you and your friend if you did the activity together and minus any activity costs. (eg \$40 happiness – \$10 costs = \$30)
- iii) Estimate (in dollar amounts) how much happiness each activity gives you and your friend if done alone and minus any activity costs.
- iv) Fill out the game theory payout matrix below. Determine which activity you will choose to do this Friday evening if your friend is allowed to choose first and decides to do activity C.

		FRIEND		
		a	b	c
YOU	a	,	,	,
	b	,	,	,
	c	,	,	,



Thank You

Next week: Externalities and Property Rights

econ1010@uq.edu.au

sam.taylor@uq.edu.au

